Dengue Weekly Hotspot & Risk Report

Target Week: 52, 2025

Area Covered: Kuala Lumpur, Malaysia

Generated By: Dengue Prediction System
Date Generated: 2025-12-14

2. Executive Summary

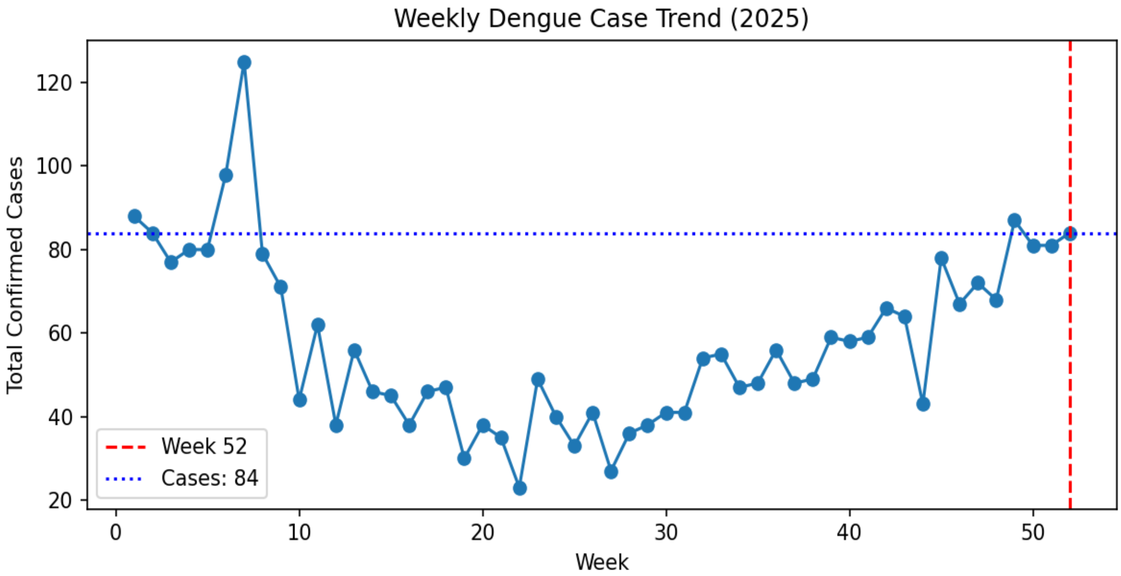
For Week 52, a total of **45** hotspot grids were identified across the scanned area. The total accumulated risk score is **92.38**, with an average grid risk of **2.05**. The highest recorded risk intensity in a single grid is **2.46**.

The report highlights critical clusters that require immediate public health attention.

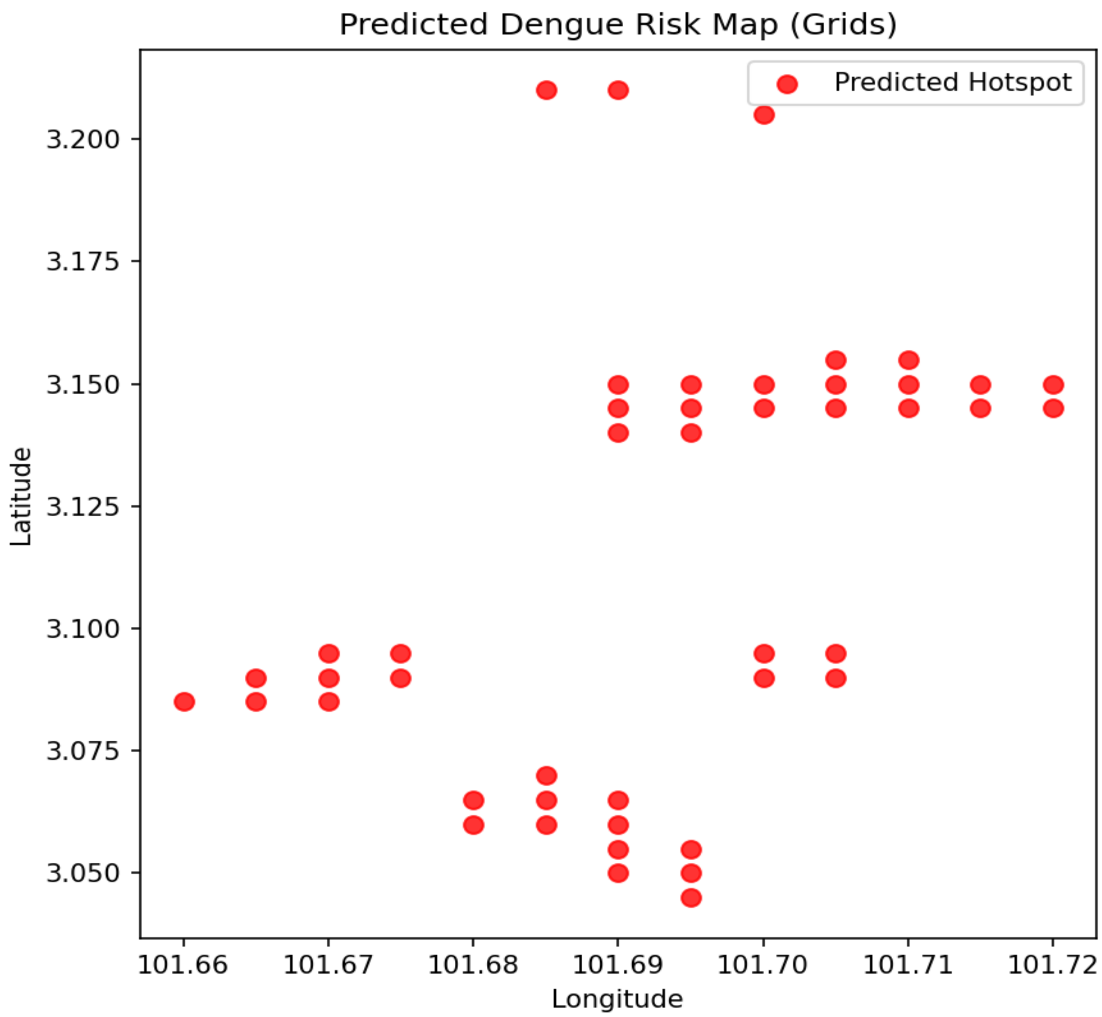
3. Data Sources & Methodology

Historical Data Source: *ultimate_combined_data.csv* containing past confirmed dengue cases.
Prediction Source: Machine-learning generated grid risk (*heatmap_data.json*).
Hotspot Definition: Any grid point with a predicted risk probability > 0 (inclusive of all detected risks).
Spatial Resolution: ~500m Grid-based latitude/longitude cells.
Time Resolution: ISO Weekly predictions.

4. Historical Dengue Trend



5. Predicted Spatial Risk Map



Predicted Dengue Risk Map (Grids)
Predicted Hotspot
3.200
3.175
3.150
3.125
3.100
3.075
3.050
Latitude
101.66
101.67
101.68
101.69
101.70
101.71
101.72
Longitude

6. Hotspot Summary Table

#	Location Area	Peak Risk	Risk Level	Grids
1	Bukit Jalil (Vista Komanwel)	2.46	MEDIUM	12
2	Desa Petaling	2.02	MEDIUM	4
3	Changkat Bukit Bintang	2.00	MEDIUM	11
4	No. 51, Jalan Alor, 50200 Kuala Lumpur	2.00	MEDIUM	1
5	Old Klang Road (Pearl Point)	2.00	MEDIUM	8
6	Pavilion Kl (Residences)	2.00	MEDIUM	6
7	Ppr Batu Muda	2.00	MEDIUM	3

7. Statistical Summary

Total Predicted Risk: 92.38
Mean Risk: 2.05
Standard Deviation: 0.15
Max Risk Score: 2.46

8. Comparison with Previous Weeks

Comparative analysis with previous week's prediction is currently unavailable in this version.

9. Risk Interpretation & Implications

Environmental Factors: High risk is often correlated with urban density and recent rainfall patterns.
Implications: Areas flagged as "High" or "Medium" risk require immediate attention for mosquito breeding site inspections.
Surveillance: Public health teams should prioritize these clusters for early intervention.

10. Recommendations

For Authorities:
- Conduct targeted fogging in the identified hotspot grids.
- Increase surveillance and larviciding in emerging danger zones.
- Launch localized public awareness campaigns.

For Public:
- Inspect home compounds for stagnant water (mosquito breeding sites).
- Use mosquito repellent and wear long-sleeved clothing.
- Seek early medical attention if experiencing fever or joint pain.

11. Limitations & Assumptions

- **Prediction Model:** Depends on historical patterns; sudden environmental shifts may not be fully captured.
- **Risk ≠ Cases:** A high risk score indicates probability, not a guaranteed outbreak.
- **Approximation:** Environmental variables are approximated for grid locations.
- **Human Movement:** The model does not currently account for daily human mobility patterns.

12. Appendix

Weight: The raw probability score from the machine learning model (scaled 0-10).
Grid Resolution: The map is divided into fine-grained cells to detect micro-clusters.
Model Accuracy: Based on training validation against historical outbreak data.